

BUNYA SECONDARY SCHOOL

S.3 CHEMISTRY TERM III HOLIDAY PACKAGE 2025

Instructions: Attempt all Items

Item 1

Fertilizers are added often to boost soil productivity. One of such is ammonium nitrate which is often imported at high costs. To boost the local production of ammonium nitrate. Government recently cleared Bloom Uganda limited set up ammonium nitrate production plant in one of the villages in Mubende district with minimal environmental impact. The locals however don't know the process of production will be carried out.

As a result, a radio talk show has been organized at one of the local radio stations to sensitize the locals.

Task: Prepare a presentation you will make during the talk show if you were invited.

Item 2: Most people in Uganda depend on agriculture for food and for employment. In addition, Agriculture contributes to the highest percentage of her export. However, the sector is faced with many challenges affecting the crops and livestock leading to poor crop and animal yields. Experts predict that if this is not solved, the country's export exchange would fall.

Task: As chemistry learner; advice the farmers on how to prevent pests and disease, (include at least 4 methods to prevent pests and for parasites).

Item 3: A group of S.3 learners came across data on melting points of elements whose identities they forgot to capture. They brought it to S.4 learners to help them analyze and interpret the data for them.

Element	X	Y	Z	W	A	Q	M
Atomic number	16	12	17	14	15	11	13
M.P/ ^o C	119	650	-101	1410	44	98	660

Task: As a senior four learner with good knowledge about periodic trends,

(a) Help them;

(i) Understand the classes of the elements.

(ii) On how to interpret the data.

(b) Suggest to them the possible uses and environmental impact of the elements.

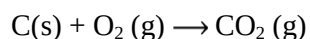
Item 4: A businessman was selling salt mixed with particles of sand, tricking people who couldn't notice the difference between the salt and its mixture with sand. This made it difficult for residents to use the salt. They need your help to prevent future problems and avoid any chemical dangers. They trust your chemistry knowledge to clarify the situation and prevent further suffering.

Task: Help the residents:

(a) Know the behavior of the substances in the mixture.

(b) Evaluate the problems associated with the use of one of the components in the mixture.

Item 5: Charcoal is a common fuel used in many homes in Uganda. During its combustion in excess oxygen supply, it produces carbon dioxide. Your friend is using 80.5g of charcoal to cook food in a busy evening for guests. It burns completely as shown in the equation below.



Task: As a learner of chemistry,

- (a) Assess the product formed.
- (b) Advise the friend on the appropriateness of the product formed.
- (c) Calculate the volume of the gas produced every busy evening.
(1 mole of a gas occupies 22.4l at s.t.p, C = 12)

Help the friend understand the impact of the product in the environment.

Item 6: A group of learners were faced with a unique solid substance, X, which they suspected to be an element.

0.3g of the element could burn in air to form 0.5g of the solid product. One of them picked interest in what could be the chemical formula of the oxide of the element. However, he did not know how to determine the formula. When they contacted the laboratory technician he gave them the atomic number and mass number of X as 12 and 24 respectively, and the symbolic representation of oxygen as ${}^{16}_8\text{O}$

Task: As a student of chemistry help the learners to;

- (a) Understand the nature of substance X.
- (b) Determine the formula of the oxide of X.
- (c) Know the environmental consequences of the element.

Item 7

Due to increasing demand of power in Uganda, the government wants to adopt use of nuclear power. However, there is little knowledge among the natives about how nuclear energy works. The ministry of energy and mineral development is interested in a knowledgeable person to sensitize the natives about the matter.

Task: As a chemistry learner,

- (a) Advise the natives on the types of processes that can be used by the government.
- (b) Educate them on how the energy can be produced and **caution** them on the possible dangers associated with the process.
- (c) Suggest to the residents on the most efficient process.

Item 8: On Kimwise's birthday party, guests were wondering and excited by the aroma and taste of the food served, that it so delicious

The Master of Ceremony (MC) kept on notifying the people that the food had been prepared with knowledge of chemistry and some of the guests wanted to know what the Master of Ceremony meant

Task: As chemistry learner, basing on the MC's communication advise the guests on:

- (a) The variety of the ingredients used.
- (b) The suitability of the ingredients.
- (c) The possible problems associated with, and choice of the ingredients used.

Item 9

Students were asked to identify examples of the changes that can take place in the atmosphere but use the same components of air. One student mentioned the burning of magnesium in air and the rusting of iron.

Task:

- (a) What component(s) of air are used during burning of magnesium and rusting of iron?
- (b) What is the similarity between the two chemical changes?
- (c) What makes rusting of iron different from burning of magnesium in air in term conditions for the reaction?
- (d) Write the formulae of the products of each of the changes.

Item 10

In your village you have challenge of water, where the wells have muddy water and the boreholes provide salt water that hardly quench their thirst but rather increase it.

Task:

- (a) Describe any scientific procedures that you as chemistry student can perform to make the village water clean and safe for human consumption
- (b) Polythene bags, car tires, buckets, iron sheets and petrol are some of the materials used in our day to day life.
 - (i) Describe any five ways how polythene bags are dangerous to our environment.
 - (ii) Explain any five ways of how to prevent the effects of polythene bags on the environment.

Item 11

James, living in an area where they use bore hole water, slid, fell and his white shirt became dirty. He decided to use a detergent to clean his shirt. The shirt remained with some brown spots yet he had rinsed it several times.

Task: As a chemistry learner;

- (a) Point out the problem James made when choosing a product.
- (b) Help James understand how the product works.
- (c) Advise James on the challenges associated with the long term use of the product.

Item 12

A resident of Matuba fetches water daily from two sources of water: borehole and nearby spring. When she uses this water from these two sources, she realized that she uses a lot of soap while washing her clothes. One day she boiled the water from the borehole for bathing but remembered that she had not washed one of her shirts and washed it with the boiled water instead and used little soap than it used to be. This has forced her to try boiling the spring water as well and used it for washing her clothes but still used a lot of soap as it used to be. This has confused her more.

Task: Prepare an article to clear away her dilemma.

Item 13

Many people in villages make a living by manufacturing local waragi (ethanol). However, the government is against the business due to associated challenges. The district's LC5 chairperson wants to sensitize the community about the large scale production process and its impact to enable them understand the government position.

The district's LC5 chairperson has contacted you as a student with the knowledge of chemistry in your village to provide the relevant information.

Task: Make a write up to use upon meeting the community.

Item 14

Salts are very important in our everyday lives as some of them are used in medicines, add taste to our food. Many of these salts can be prepared in the laboratory.

Task:

- a) Suggest three ways in which salts can be prepared in the laboratory school.
- b) Ammonium chloride is an example of the salts prepared from the laboratory. When it was dissolved in water and the solution tested, it was found to be acidic.
 - (i) Explain why the solution was acidic.
 - (ii) State three other applications of salts in your community.

Item 15

Large deposits of oil have been discovered in Lake Albert near Hoima city. Crude oil contains a variety of different compounds but with similar structure.

The simplest of these compounds is methane of molecular formula CH_4 and many others.

Task:

- a) Identify the class and write the general molecular formula of the organic family to which the components of oil belong.
- b) Name and write the structural formula of the next two members of the organic family named after methane.
- c) Giving a reason for your answer, state one method by which the components of crude oil.

Item 16

The chairperson of Matyama village in Namutumba district has banned residents from using charcoal as their homesteads. The idea was to reduce deforestation trends in the region. She has heard of biogas as a better alternative and would like to set up a biogas in production unit at her residence as a demonstration point for the community but with minimum environmental impact. Many of the residents have less interest in this unit and some have gone beyond saying it will be of no use to their community. She has contacted you as an educated member of the community to sensitize the residents in a community meeting on how the process of production can be carried out.

Task: As a chemistry student, make a write up you will use during plenary.

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We Wish You A Merry Christmas And A Happy New Year - 2026